

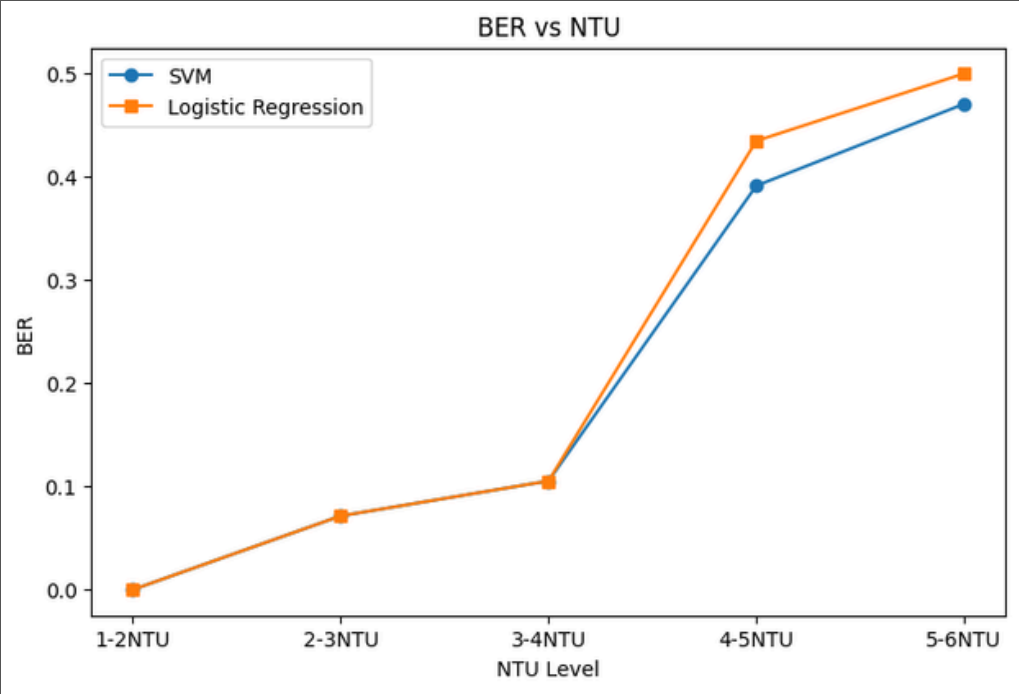
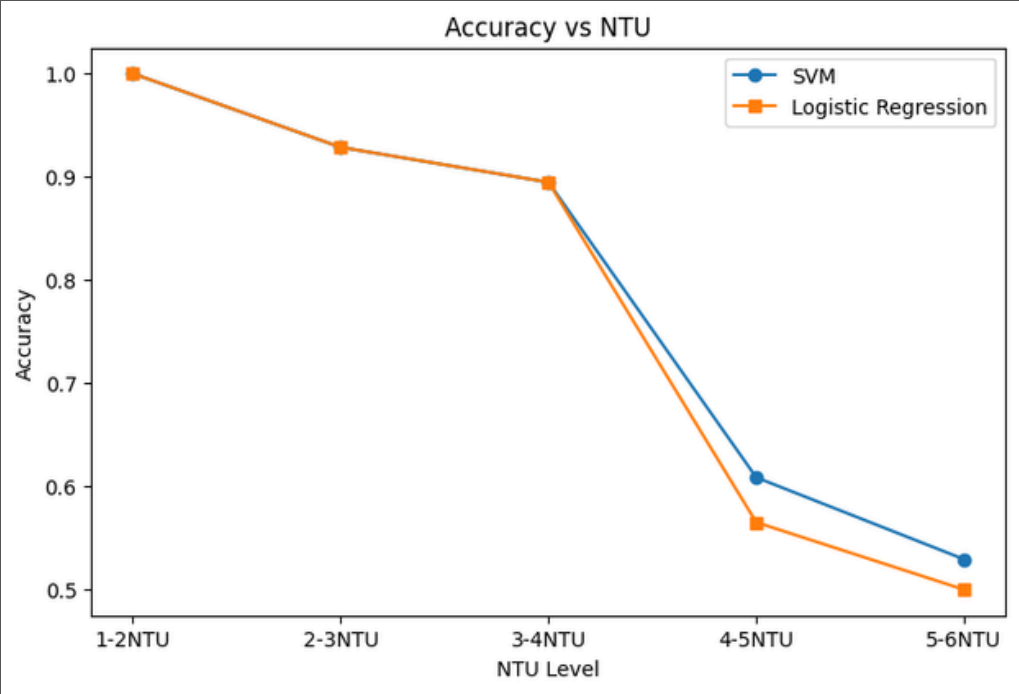
Day 20 - Robustness Study

SVM NTU-wise Results :

	NTU	Accuracy	Precision	Recall	F1	BER
0	1-2NTU	1.000000	1.000000	1.0	1.000	0.000000
1	2-3NTU	0.928571	0.818182	1.0	0.900	0.071429
2	3-4NTU	0.894737	0.777778	1.0	0.875	0.105263
3	4-5NTU	0.608696	0.000000	0.0	0.000	0.391304
4	5-6NTU	0.529412	0.000000	0.0	0.000	0.470588

Logistic Regression NTU-wise Results :

	NTU	Accuracy	Precision	Recall	F1	BER
0	1-2NTU	1.000000	1.000000	1.0	1.000000	0.000000
1	2-3NTU	0.928571	0.818182	1.0	0.900000	0.071429
2	3-4NTU	0.894737	0.777778	1.0	0.875000	0.105263
3	4-5NTU	0.565217	0.000000	0.0	0.000000	0.434783
4	5-6NTU	0.500000	0.470588	0.5	0.484848	0.500000



Research Insight :

Effect of Turbidity (NTU)

- Low NTU conditions (1-4 NTU) provide clear signal separation between transmitted bits.
- As NTU increases, scattering and absorption distort the received optical signal.
- Signal overlap between bit 0 and bit 1 increases.
- Classification becomes more difficult at higher turbidity levels.
- BER increases while Accuracy and F1-score decrease.

